Mental Models in Conceptual Development

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Abstract: Models are important in science because they can be used as instruments to help in the construction of theories. They are the sources of both predictive and explanatory power and can be used to suggest new hypotheses and aid in scientific discovery. In this paper I argue that the mental models that children construct also have predictive and explanatory power and can be used as mediating mechanisms for the revision of existing theories and the construction of new ones.

1. INTRODUCTION

1.1 Mental model as sources of predictive and explanatory power

For a number of years now, I have been engaged in research attempting to understand the development of knowledge about the physical world and the learning of science concepts. In this paper, I will focus on one aspect of this research that has to do with the role of mental models in conceptual development and in reasoning about the physical world. More specifically, I will argue that the ability to form mental models is a basic characteristic of the human cognitive system and that the use of models by children is the foundation of the more elaborate and intentional use of models by scientists.

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Models are important in science because they can be used as instruments to help in the construction of theories. They are the sources of both predictive and explanatory power and can be used to suggest new hypotheses and aid in scientific discovery. Similarly, I will argue that the mental models that children and lay adults construct are important for conceptual development and conceptual change. They also have predictive and explanatory power and can be used as mediating mechanisms for the revision of existing theories and the construction of new ones.

The paper starts with a discussion of the theoretical framework within which conceptual development is considered. More specifically, it is argued that concepts are embedded within framework and specific theories and that conceptual development involves not only the enrichment but also the revision and radical restructuring of these theories. Mental models are defined as a form of mental representation that preserves the structure of that which it represents. It is argued that mental models can be particularly useful in situations where implicit physical knowledge needs to be exploited for the purpose of answering a question, solving a problem, or in order to understand incoming information. Examples are given of how mental models function in conceptual development and how they are used in the construction and revision of theories about the physical world.

1.2 Concepts are embedded in framework and specific theories

Cognitive developmental research has so far confirmed the hypothesis that the knowledge acquisition process starts at birth and that infants proceed rapidly to construct certain fundamental understandings of the physical and social world (e.g., Carey, 1985; Wellman and Gelman, 1992). These fundamental understandings appear to develop in at least two core domains, physics and psychology with biology emerging as a possible third domain (Inagagi and Hatano, 1996).

Many developmental psychologists describe concepts in these core domains as entrenched in naïve or folk theories (although this issue is still debated especially outside developmental psychology). The term "theory" is used here relatively freely to distinguish a coherent body of knowledge that involves causal, explanatory understanding, as compared to a body of knowledge that involves a collection of facts and some procedures for operating on them. It is assumed that children's theories are different from scientific theories because children's theories lack many important characteristics of scientific theories, such as, their systematicity, abstractness, and social/institutional nature. In addition, it appears that children lack metaconceptual awareness of their theories, do not make the explicit distinction be-

tween theory and evidence, and do not understand how theories guide the hypothesis testing process (Carey and Smith, 1993; Kuhn, Amsel, and O' Longhlin, 1988).

Some developmental psychologists make the distinction between *framework theories* and *specific theories* following a similar distinction in the history of science between paradigms (research programs, or research traditions) and specific theories (Kuhn, 1962; Lakatos, 1970; Laudan, 1977). In the area of physics, a framework theory would include the basic ontological and epistemological presuppositions that define the concept of the physical object. Such a framework theory of physics would be different, for example, from a framework theory of psychology both in its ontology (physical objects versus psychological beings) and in its causality (mechanical causality versus intentional causality).

Indeed it appears that young children distinguish physical objects from psychological beings from early on. Research with infants has documented the rapid acquisition of a coherent body of physical knowledge concerning primarily certain types of principles, or presuppositions, that further define the ontology of the physical object as well as the notion of mechanical causality. For example, using preferential looking methods, Spelke and her colleagues have shown that 3-month-old infants will perceive three dimensional objects presented against a uniform background as connected bodies that will maintain their cohesion as they move (Spelke, Breinlinger, Makomber, and Jacobson, 1992). Other experiments suggest that 4-month-old infants understand that surfaces move together only if they are in contact and independently only if they are separated (Kellman and Spelke, 1983; Kellman, Spelke and Short, 1986). Infants provide evidence that they start to understand that objects move downward in the absence of support (gravity principle) or continue to move in the absence of obstacles (principle of inertia) between 6 and 8 months (Spelke, 1991; Baillargeon, 1990). In addition, experiments by Leslie on 6-month-old infants' perception of physical causality suggest an early appreciation of some aspects of mechanical causality (Leslie and Keeble, 1987; Leslie, 1988), although, according to Oakes and Cohen (1995), physical causality does not appear until the age of 10 months. Certainly by the end of the first year, infants understand how physical causation works and appreciate the differences between psychological and physical causality (Meltzoff, 1995).

Specific theories are embedded within framework theories and are constrained by them. For example, research with preschool and elementary school children has shown that young children categorize the earth as a physical object (rather than as an astronomical object) and apply to it all the properties of physical objects. The specific theory of the earth is constructed on the basis of children's observations and cultural information they receive

in their everyday life under the constrains of the framework theory of physics (Vosniadou, 1994a; Vosniadou and Brewer, 1992). Figure 1 shows the hypothetical conceptual structure underlying children's initial concept of the earth.

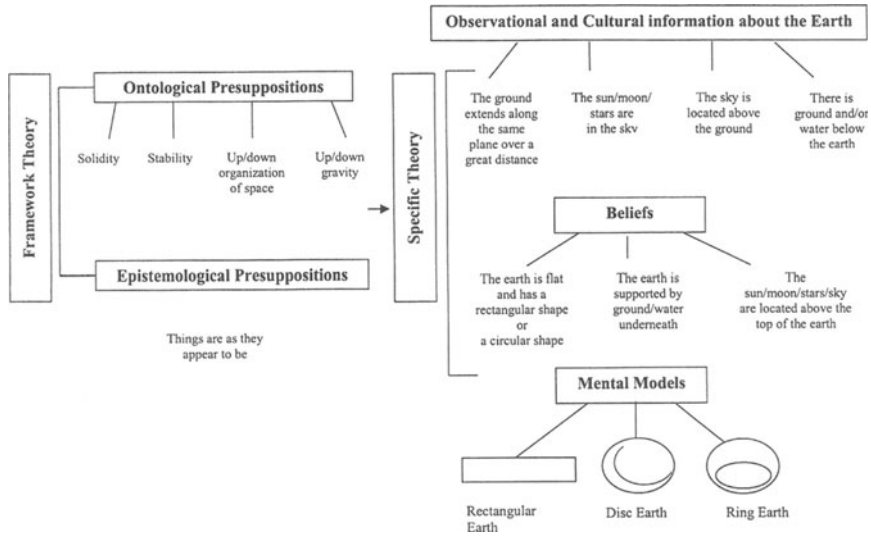


Figure 1. Hypothetical conceptual structure underlying children's initial mental models of the Earth.

1.3 Mental models

In order to give a full account of conceptual development, another theoretical construct appears to be needed, that of the *mental model.* Mental models are analog representations that preserve the structure of the thing they represent. It is assumed that most mental models are constructed on the spot to deal with the demands of specific situations, although it is possible that some mental models may be stored in long-term memory. When a representation in the form of a mental model is constructed, this representation is designed to be as useful as possible in the situation in which it is involved (in the solution of problem, to answer a question, etc.). However, mental models are also constrained by the framework and specific theories within which they are embedded and thus can be important sources of information about them (see Vosniadou, 1994a).

An important characteristic of mental models is that they can be explored extensively, run in the mind's eye, so to speak, in order to generate predic-

tions and explanations. This point will be developed later in greater detail. Experiments in cognitive psychology have shown the psychological reality of mental models (e.g., Sheppard and Metzler, 1971; Kosslyn, 1994), while experiments in cognitive neuroscience have provided neurological evidence for mental models.

In examining the role of mental models in conceptual development and conceptual change I will focus on two aspects: The first has to do with the *construction of mental models* and the *second with their function*.

2. THE CONSTRUCTION OF MENTAL MODELS

Figure 2 shows the mental models of the earth constructed by elementary school children in a study designed to investigate children's understanding of the shape of the earth (Vosniadou and Brewer, 1992).

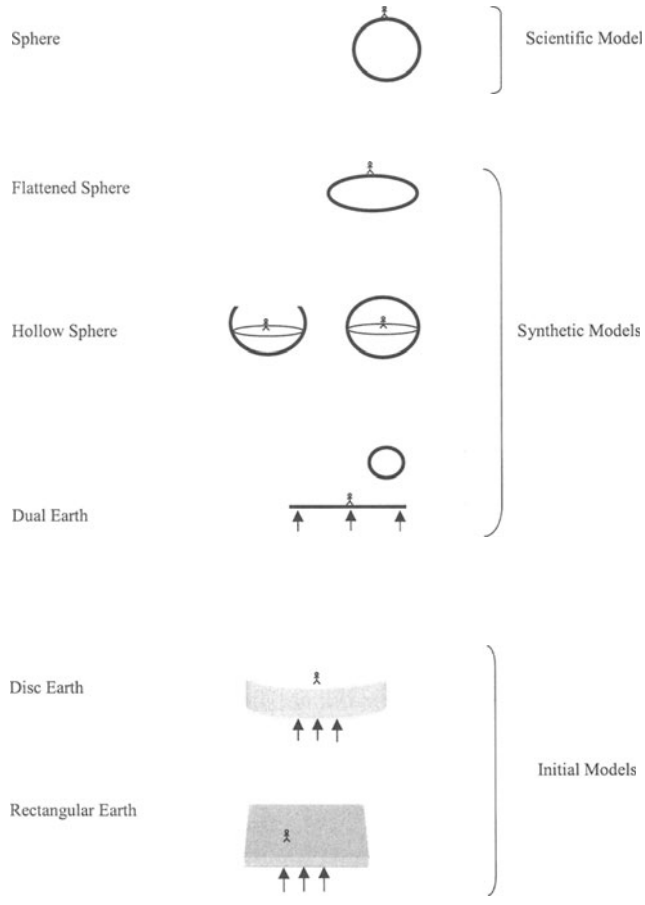


Figure 2. Mental models of the Earth.

These mental models were obtained by combining data coming from children's answers to verbal questions with drawings and models made out of play-dough. The children were asked a number of questions about the shape of the earth and where people live on the earth. They were also asked to make drawings and play-dough models of the earth.

One important aspect of the methodology was the use of generative questions. Generative questions cannot be answered on the basis of stored information but require the genuine solution of a new problem. For example, the question "If you were to walk for many days, would you ever reach the end of the earth? Does the earth have an end/edge?" is assumed to be a generative question. This is because most children have not encountered this question before and therefore do not have explicit conceptual knowledge from which they can derive ready-made answer. It is assumed that in order to answer a generative question, our subjects must create a mental representation or a mental model of the earth, and explore it in order to derive from it a relevant answer.

During the data analysis we try to determine whether each individual subject's responses (both verbal and non-verbal) form a coherent pattern consistent with the use of a generic model of the earth. For example, if a given subject draws a flat, rectangularly-shaped earth, creates a similar play-dough model, answers that the earth is flat, and that people could fall off the flat end/edge of the earth, this is taken as adequate evidence to support the claim that this subject is using a model of a flat earth to reason from.

Our studies in the United States (Vosniadou and Brewer, 1992; 1994) but also a number of cross-cultural studies in Greece (Vosniadou et al., 1996), in India (Samarapungavan, Vosniadou and Brewer, 1992), and in Samoa (Vosniadou, 1994b) have shown that a) there is a small number of overall models of the earth that our subjects form, b) about 80-85% of the subjects in our sample use one of these models in a relatively consistent way, and c) there are interesting cross-cultural similarities and differences in the use of these models.

Recently, a study by Ioannidou and Vosmiadou (2001), following a similar methodology investigated children's and adults representations of the layering and composition of the inside of the earth and obtained similar findings. Figure 3 shows the models of the layering and composition of the earth that the subjects of this study formed.

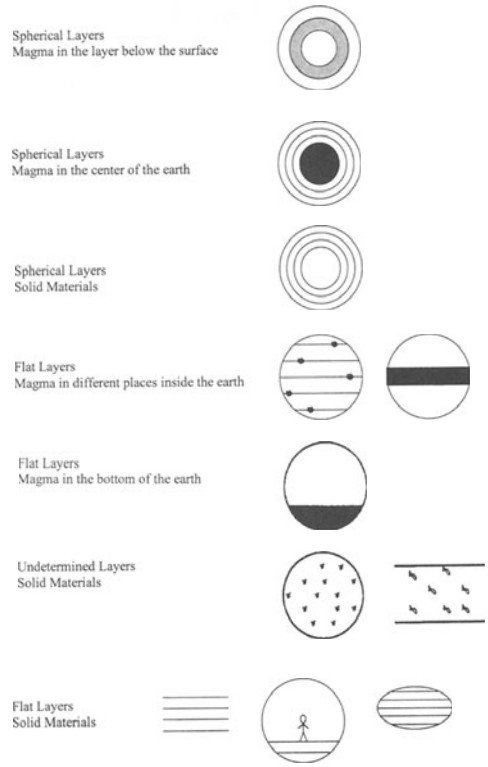


Figure 3. Models of the layering and composition of the earth.

3. HOW MENTAL MODELS FUNCTION

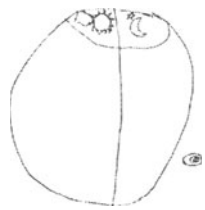
Mental models can have various functions in the human cognitive system. Here we will discuss three important functions of mental models: a) as aids in the construction of explanations, b) as mediators in the interpretation and acquisition of new information, and c) as tools that allow experimentation and theory revision.

3.1 Mental models as aids in the construction of explanations

When people reason about the physical world they often make use of mental models, particularly in situations where the answer cannot be retrieved from previously stored information or can be easily deduced from verbal information. Mental models can help people to draw on implicit physical knowledge that can be then used to answer questions, solve prob-

lems, etc. By doing so, the implicit knowledge becomes explicit and conceptual.

It was mentioned earlier that one important aspect of the methodology used in the astronomy studies was the use of generative questions that we believe stimulates the formation of a mental model. Mental models are helpful in these cases precisely because the generative questions cannot be answered on the basis of previously known facts or explanations. In these situations, people tend to draw on their implicit physical knowledge in an effort to find relevant information. The construction of a mental model allows people to represent the relevant physical knowledge in a way that can help them solve the problem at hand.



E: So what is the real shape of the earth?
C: Round.
E: Why does it look flat?
C: Because you are inside the earth.
E: If you walked and walked for many days in a straight line, where would you end up?
C: Somewhere in the desert.
E: What if you kept walking?
C: You can go to states and cities.
E: What if you kept on walking?
C: (No response).
E: Would you ever reach the edge of the earth?
C: No. You would have to be in a spaceship if you're going to go to the end of the earth.
E: Is there an edge to the earth?
C: No. Only if you go up.

Later:
E: Can people fall off the end/edge of the earth?
C: No.
E: Why wouldn't they fall off?
C: Because they are inside the earth.
E: What do you mean inside?
C: They don't fall, they have sidewalks, things down like on the bottom.
E: Is the earth round like a ball or round like a thick pancake?
C: Round like a ball.
E: When you say that they live inside the earth, do you mean they live inside the ball?
C: Inside the ball. In the middle of it.

Figure 4. Answers to the question regarding the earth end/edge from the point of view of a child with a mental model of the earth as a hollow sphere.

The example shown in Figure 4 demonstrates how one of our subjects, Venica, an 8- year old child, imagines the earth as a hollow pumpkin open at the very top where the various solar objects are drawn, and derives from this model a surprising answer to our question regarding the end/edge of the earth. More specifically, Venica believes that people live inside the earth, at the bottom. From her point of view this hollow earth is open at the very top, where the solar objects are located, but this top is very high up and you need a spaceship to get there. Her answer that "you would have to be in a spaceship if you are going to go to the end of the earth", shows how her mental model of the earth leads her to provide this interesting answer to our question about the end/edge of the earth.

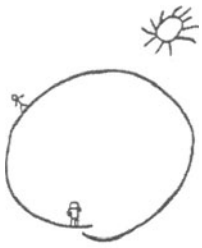
What is important to mention here is that the mental model appears to mediate in the generation of the explanation. The mental model becomes the vehicle through which implicit physical knowledge enters the conceptual system. Once this implicit physical knowledge has entered the conceptual system, it can become explicitly coded and thus available for further theorizing. It is in this respect that mental models can become important sources for the generation of new conceptual knowledge, and can aid in theory construction.

3.2 Mental models as mediators in the interpretation of new information and in learning

Mental models can exert important influence on the interpretation of new information. In the examples discussed below, we can see how children's mental models of the earth influence the way information coming from an outside source is interpreted. These examples are taken from a study investigating the development of children's explanations of the day/night cycle (Vosniadou and Brewer, 1994). In this study we presented children with the drawing shown in Figure 5. We told them that this is a drawing showing a person on the earth and that we wanted them to "make it so it is day for that person" and then to "make it night".

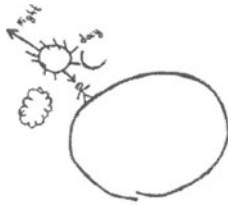


Figure 5. Drawing given to the children in order to explain the day/night cycle.



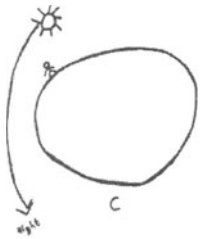
Drawing No. 1

Tamara (No. 9, Grade 5)
The sun is occluded by clouds or darkness
E: Now can you make it so it is day for that person?
C: He's outside the earth.
E: Where should he be?
C: In here (see Figure 4, drawing 1)
E: …OK now, make it daytime for him.
C: The sun is out here, but it looks like it's in the earth, when it shines…
E: OK. What happens at night?
C: The clouds covered it up.
E: Tell me once more how it happens.
C: Cause at 12 o'clock it's dark.



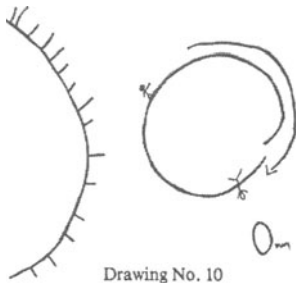
Drawing No. 3

Allison (No. 52, Grade 1)
The sun moves out into space.
E: Now make it so it is day for that person.
C: (child makes drawing 3 shown in Figure 4) Right here?
E: Whatever you think. Now make it night.
C: It goes in space.
E: Show me. Tell me how it happens.
C: The sun comes back down. It goes into space and when it gets dark the moon comes back out.



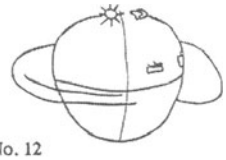
Drawing No. 5

Timothy (No. 47, Grade 1)
The sun goes down to the other side of the earth (and the moon goes up)
The child makes the drawings shown in Figure 4.
E: Tell me once more how it happens.
C: When the moon comes up and the sun goes down.
E: Where was the moon before?
C: Under the earth.
E: Show me. Tell me how it happens.
C: What time was it when it goes under the earth?
C: Day



Drawing No. 10

Robert (No. 5, Grade 5)
The earth rotates up/down and the sun and moon are fixed at opposite sides.
E: Now make it so it is day for that person.
C: (child makes drawing 10 shown in Figure 4).
E: Now can you make it night time?
C: Can I draw him somewhere else? (draws figure at the bottom of the earth)
E: Sure
C: (child draws arrows to show how earth spins)
E: Tell me how it happens.
C: When it was daytime, the earth spinned around to the sun. When it was night time, the earth turned around to where the moon is.



Drawing No. 12

Venica (No. 33, Grade 3)
The earth rotates left/right and the sun and moon are fixed at opposite sides.
C: (child makes drawing 12, Figure 4).
E: Can you tell me how this happens?
C: When the earth turns around its orbit, this side comes day and the other side comes night.

Figure 6. Examples from children's explanations of the day/night cycle (cf. also the previous page).

As can be seen from the examples shown in Figure 6, the children interpreted this drawing in different ways. For example, Tamara (Drawing No. 1) thought that our drawing was wrong because the person was "outside the earth". When we told her to show us where the person should be, she drew the person inside the earth at the bottom and explained the day/night cycle in terms of the sun being covered by clouds.

Allison (Drawing No. 3) accepted our drawing and added the sun to make it day, explaining that the sun "goes in space" and "when it gets dark the moon comes back in". Timothy also accepted our drawing, but he had a very different model of the day/night than Allison. He thought that the sun goes down to the other side of the earth (Drawing No. 5).

Compared to all of the above, Drawing No 10 by Robert is very different. It shows a huge sun, bigger than the earth, and a demonstration of the earth's rotation in an up/down fashion, different from the east/west rotation of the earth envisioned by Venica (Drawing, 12).

It should be mentioned here that children's interpretations of our drawing are consistent with their models of the earth and their beliefs about where people live on the earth. Our studies have shown that children's mental models of the earth act as strong constraints on the way children interpret infor-

mation regarding the day/night cycle. There was not even a child in our sample that provided interpretations of the day/night cycle based on the earth's axis rotation or revolution around the sun that had not understood the spherical shape of the earth.

It is interesting to note here that most of the children who had a spherical shape of the earth and attributed the day/night cycle to be the earth's axis rotation interpreted this rotation to be an up/down and not an east/west rotation. We think that they do so because the belief in the up/down rotation of the earth is consistent with the phenomenal appearance of the sun to be "up in the sky" as well as with children's pre-scientific explanations of the day/night cycle in terms of the sun going down behind the mountains or at the other side of the earth.

In contrast to the children who have understood that the earth is a rotating sphere, most of the children who form hollow sphere models, like Venica, interpret the rotation to be an east/west rotation, not an up/down one. Again, this makes sense in view of the fact that the hollow-earth children believe in "up/down gravity" and think that the people live inside the earth because they would fell down if they lived outside. For those children it does not make sense to consider the earth's axis rotation to be an up/down one, because then the people would fall down every time the earth turned.

All of the above can be used as arguments for the psychological reality of mental models. It appears that on the one hand mental models are constrained by prior beliefs and presuppositions and on the other, once formed, they themselves act as constraints on the way the new information that enters the conceptual system is interpreted.

3.3 Mental models as aids in experimentation and theory revision

In order to show how mental models can aid in experimentation and theory revision I will use two examples. The first is from an unpublished astronomy study I did in Greece with adult illiterates. This was a population of relatively young men and women who for some reason or other had dropped out of school after the third or fourth grade. The women were housewives and the men were construction workers or truck drivers. The man I will talk about was about 30 years old, a truck driver, who was very interested in the interview and said that the questions I was asking him were often the subject of conversations with his friends in the cafes at night after work. He believed the earth to have the shape of an egg and thought that people lived only on the "top part of the earth and not at the bottom" because they would fall.

He made a drawing of the egg-shaped earth on a piece of paper and drew a person standing at the top of this earth. At a later part of the interview I

asked him if the earth moved. He said that he knew that the earth rotated and, like most people, interpreted the rotation to be an up/down rotation. When he said that, he made an up/down movement using his hands in order to illustrate the movement of the earth. As he was doing this he came to realize the obvious inference that if the earth rotated up/down then the people who lived at the top of the earth would eventually go down. At this time he stopped and he said that there was something wrong here.

He was absolutely sure that the earth moved but he was also sure that the people could not live at the "bottom" of the earth! Since he could not solve this apparent inconsistency I told him not to worry and we proceeded with the interview. When the interview was almost over, he stopped me and said that he had found a solution to the inconsistency regarding the earth's movement. He said that the earth moved all right but what moved was only the "core" of the earth and not the whole earth. Since only the core of the earth moved up/down, the people on the outside of the earth could go on living undisturbed.

Misconceptions such as the one discussed above, are often creative solutions to the problem of incorporating theoretically inconsistent information into the knowledge base. They demonstrate the constructive nature of human cognition, and are important mediators in the process of conceptual change.

The second example I will use comes from my younger daughter Irene when at the age of four served as a pilot subject for the astronomy studies, At that time, I was in Greece, training a group of students who were going to go to a school to test children. I had brought my daughter and an older child (8 years of age) to be the pilot subjects. We started with Irene who gave consistently flat earth responses to all questions, although she had been exposed to some spherical earth information. When she finished we examined the second child who gave consistently spherical earth responses to the questionnaire. Irene was listening to the other child carefully and at the end of the interview she said to us: "I want to be asked again". We were all surprised by her request but we agreed to have her tested again. So we started asking her the questions from the beginning. One of the first questions in the questionnaire was the question "What is the shape of the earth?" and "Take this play-dough and make a model of the earth that shows its real shape". Irene took the play-dough and with a very happy smile made a spherical earth for us. We continued with the remaining questions one of which involved showing to the subject the picture of a farm-house on what appeared to be a flat farm land. In this question the usual procedure was to tell the subject: "This is a picture of a house on the earth. The earth appears to be flat here. Do you agree?" (Most children would agree). Then we would say: "Can you explain to us why the earth appears to be flat here when you said before that the earth was round?" This question was asked only if the subject had said

that the earth is round. Irene looked at us and thought for a while how to answer the question. She then took the spherical earth that she had made with the play-dough and pressed it in her hands flattening it and turning it into a flat pancake, looking again very happy at her solution.

In both of these examples the mental model appears to mediate in the interpretation of information that comes from the outside and plays an important role in how this information is used for theory revision.

4. THE AUTONOMY OF MENTAL MODELS

In their recent book *Models as Mediators* Morgan and Morrison (1999), argue that the reason that models can function as important instruments in science is because they occupy an autonomous role, they are *autonomous agents*. "It is because they are neither one thing nor the other, neither just theory nor data, but typically involve some of both – that they can mediate between theory and the world". I would like to suggest that something similar also applies to mental models. In other words, mental models play an important role in conceptual development and conceptual change, because they are not, completely, determined neither by data nor by theory. They retain a degree of independence that is responsible for the fact that they can function in ways that promote theory development.

Is there any empirical evidence in support of this claim? I think that such evidence can be found in the considerable individual and cultural differences in the mental models that exhibit similar constraints. For example, there are a number of different models of the earth that all express the flat ground and up/down gravity constraints, such as the model of the earth as a round pancake/disk, as a flattened sphere or as an egg-shaped sphere with the people living only on the top part, or the model of the earth as a hollow sphere with people living on flat ground inside it. Furthermore, there is evidence that there are systematic cross-cultural differences in the way children interpret the constraint that the earth itself is supported by something. Thus, children from India appear to prefer an earth supported on an ocean of water, an idea not very popular with American children, who think the earth is supported by ground all the way down, or with Greek children, some of whom think that the earth rests on the strong shoulders of Atlas! (Vosniadou, 1994a).

The fact that the same constraints can be embodied in different and distinct mental models is significant if we consider (as was previously argued) that the mental model is the basis on which new information enters the conceptual system. Thus, a child with the model of a disk earth, or a flattened sphere, can explain the day/night cycle in terms of the sun going down to the other side of the earth, but this explanation does not make sense from the point of view of Venica who drew the model of the hollow sphere shown in figure 6, drawing 12. Potentially, such differences in mental models of the

earth, regardless of the fact that they started as embodiments of the same theoretical constraints, may give rise to completely different theories.

CONCLUSION

I have argued that the ability to form mental models is a basic characteristic of the human cognitive system and that the use of models by children is the foundation of the more elaborate and intentional use of models by scientists. The mental models that children and lay adults construct have predictive and explanatory power and can be used as mediating mechanisms for the revision of existing theories and the construction of new ones. This is the reason why they are important in conceptual development and conceptual change.

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